Rotations of the circle form an abelian group under composition

The Open Logic Project contributors

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Editorial scope. Single lemma card cut from the Open Logic Project unit olp-set-theory-choice (pinned-original-source.tex); statement and proof are the author's own text at source lines 936-938 and 940-956.

Lemma 0.1. SO(2) *forms an abelian group under composition of functions.*

Proof. Writing $0_{\mathrm{SO}(2)}$ for the rotation by 0 radians, this is an identity element for SO(2), since $\circ 0_{\mathrm{SO}(2)}\rho = \circ\rho 0_{\mathrm{SO}(2)} = \rho$ for any $\rho \in \mathrm{SO}(2)$.

Every element has an inverse. Where $\rho \in \mathrm{SO}(2)$ rotates by r radians, $\rho^{-1} \in \mathrm{SO}(2)$ rotates by $2\pi - r$ radians, so that $\rho \circ \rho^{-1} = 0_{\mathrm{SO}}(2)$.

Composition is associative: $\circ\rho(\circ\sigma\tau) = \circ(\circ\rho\sigma)\tau$ for any $\rho, \sigma, \tau \in \mathrm{SO}(2)$

Composition is commutative: $\circ\rho\sigma = \circ\sigma\rho$ for any $\rho, \sigma \in \mathrm{SO}(2)$. □